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ALM – PROTOTYPE REVIEW

submitted by

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Abstract

This report presents two educational cybersecurity projects developed for the Information Assurance and Security course: **SecureTunnel – Security Protocol Visualizer** and **SENTINEL – Intrusion Detection and Firewall Simulator**. The main objective of these projects is to provide an interactive learning environment for understanding secure communication protocols, network protection mechanisms, and intrusion detection techniques.

SecureTunnel visualizes three important security protocols: Transport Layer Security (TLS), Secure Shell (SSH), and Internet Protocol Security (IPsec). It demonstrates handshake procedures, authentication concepts, and packet protection. The TLS module explains secure connection establishment, the SSH module illustrates secure remote connection procedures, and the IPsec module explains Encapsulating Security Payload (ESP), including Transport Mode and Tunnel Mode. These visualizations help users understand how security protocols protect data during network communication.

SENTINEL focuses on network perimeter security through simulated firewall rule evaluation, intrusion detection, statistical anomaly detection, and bastion-host access verification. Its firewall module evaluates simulated packets against configured rules to determine whether traffic should be allowed or blocked. The intrusion detection module identifies predefined suspicious patterns, including repeated failed SSH login attempts, Telnet connections, port scanning, and excessive traffic. Statistical anomaly detection compares traffic measurements with a historical baseline to identify unusual behaviour.

Together, they provide a practical foundation for learning essential cybersecurity principles.

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Chapter 1

Introduction

1.1 Project 1: SecureTunnel

SecureTunnel is a Security Protocol Visualizer. It helps students understand how secure communication is established and how network traffic can be protected. Its main modules include TLS Handshake, SSH Connection, IPsec ESP, protocol comparison, and learning resources.

1.2 Project 2: SENTINEL

SENTINEL is an Intrusion Detection and Firewall Simulator. It demonstrates rule-based packet filtering, signature-based IDS alerts, statistical anomaly detection, and perimeter access through a simulated bastion host. It provides a dashboard for observing simulated traffic and security decisions.

1.3 Objectives

- Visualize important network security protocols in an understandable way.
- Demonstrate firewall rule evaluation and packet allow/block decisions.
- Illustrate common IDS signatures and statistical traffic anomalies.
- Explain the role of a bastion host in restricting administrative access.
- Connect classroom security concepts with interactive web applications.

Chapter 2

Technology Stack and Architecture

2.1 SecureTunnel

The documented stack includes React, Vite, Tailwind CSS, Axios, Python, FastAPI, Pydantic, and the Python `cryptography` library. The frontend sends JSON requests to backend APIs and displays returned protocol steps and demonstration results.

2.2 SENTINEL

The documented stack includes React with Vite, React Router, Recharts, Lucide React, Python 3.12, FastAPI, SQLAlchemy, SQLite, NumPy, Pydantic, pytest, and HTTPX. Its architecture separates the frontend, REST API routers, firewall and detection engines, and SQLite storage.

Chapter 3

SecureTunnel: Modules and Interface

3.1 Home Page

The home page introduces SecureTunnel and provides navigation to the TLS, SSH, IPsec, protocol comparison, and learning sections.

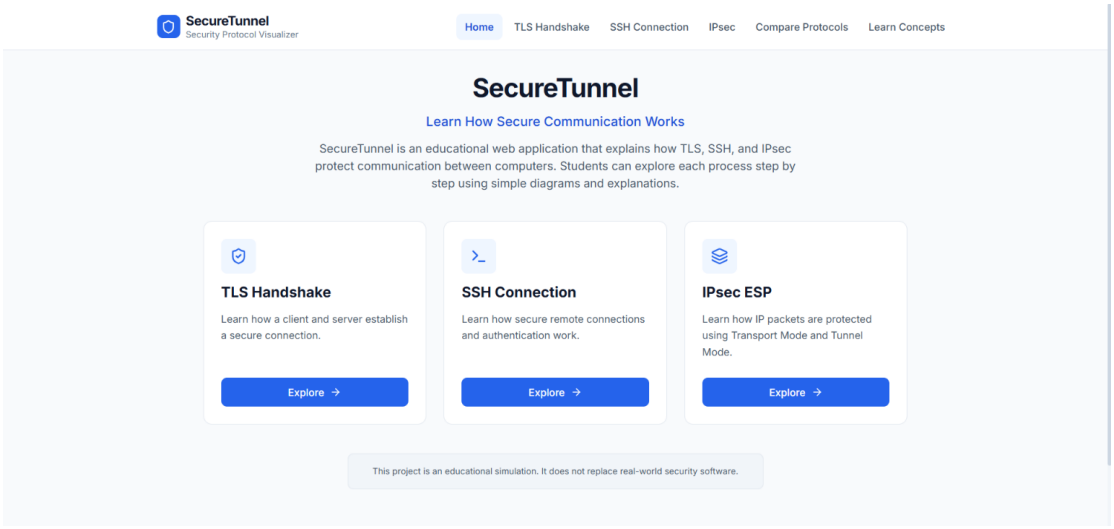


Figure 3.1: SecureTunnel home page

3.2 TLS Handshake Visualizer

The TLS module demonstrates selected handshake steps using TLS 1.2 RSA and TLS 1.3 Diffie–Hellman teaching modes. Users can start the handshake and inspect explanations of how a client and server establish key material. The interface is an educational simulation, not a real HTTPS connection.

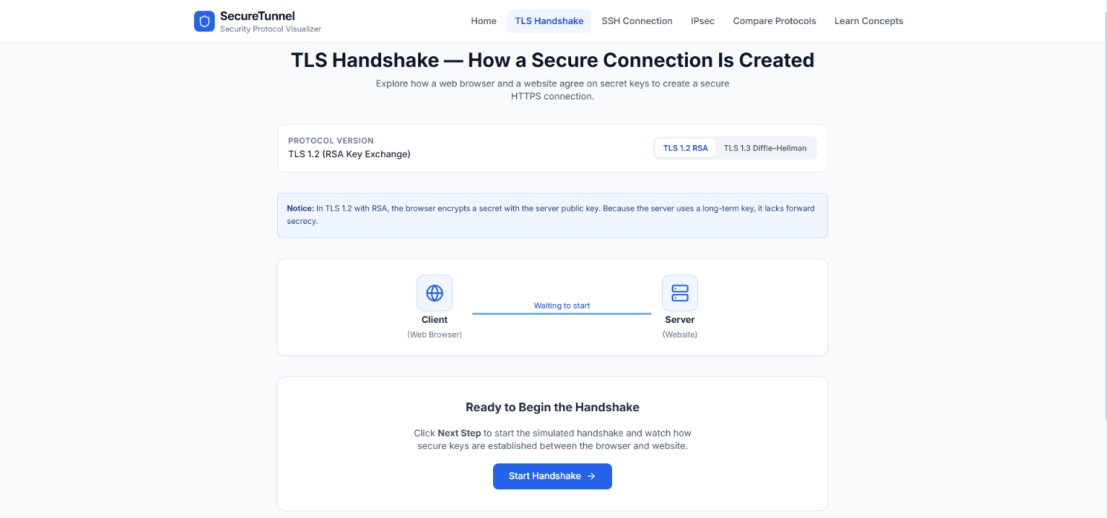


Figure 3.2: TLS handshake visualization

3.3 SSH Connection Visualizer

The SSH module illustrates secure remote connection setup and distinguishes server authentication from user authentication. Password-based and key-based scenarios are simulated; displayed command output is not produced by a real remote shell.

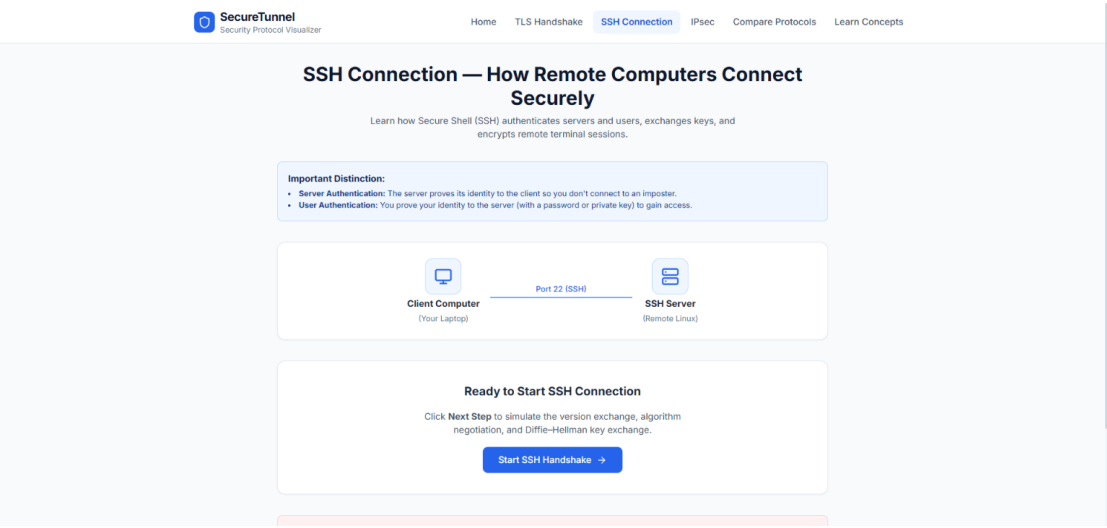


Figure 3.3: SSH connection visualization

3.4 IPsec ESP Visualizer

The IPsec module explains Encapsulating Security Payload (ESP) and lets users compare Transport Mode and Tunnel Mode. The interface displays the original packet and a simulated protected packet.

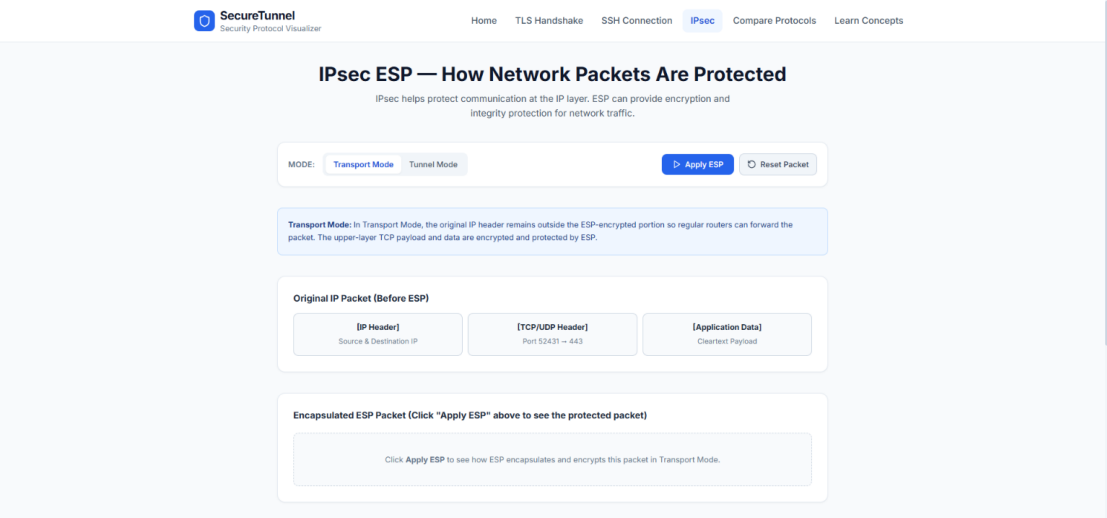


Figure 3.4: IPsec ESP packet-protection interface

Chapter 4

SENTINEL: Modules and Interface

4.1 Security Operations Overview

The dashboard summarizes simulated packet totals, allowed and blocked decisions, IDS alerts, high/critical alerts, and anomaly status. It also shows traffic trends and a breakdown of firewall decisions.

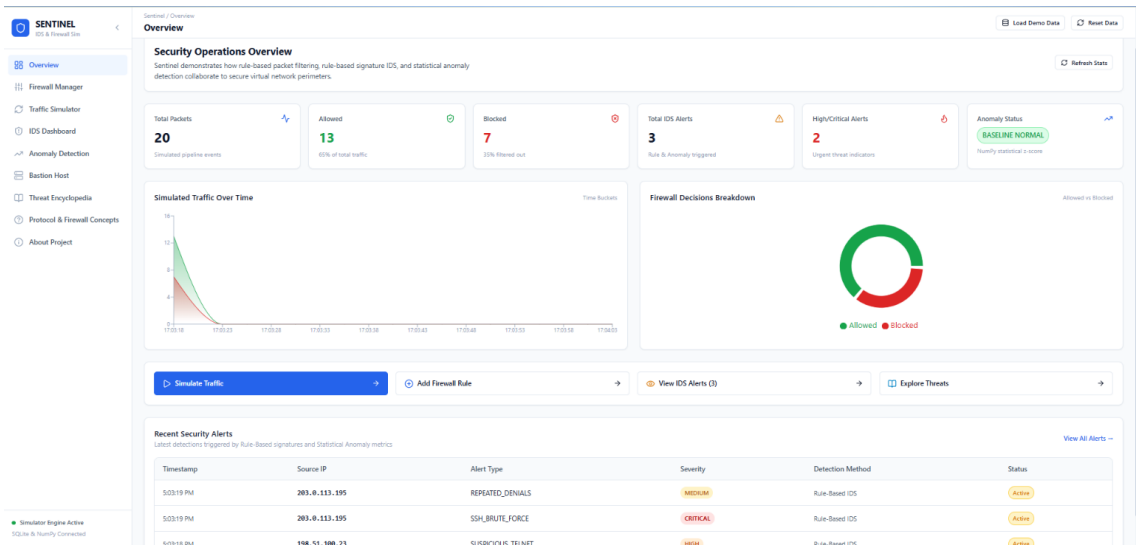


Figure 4.1: SENTINEL security operations overview

4.2 Firewall Manager

The firewall manager displays enabled rules in priority order. The engine evaluates source IP, destination IP, protocol, and destination port. The first matching enabled rule determines whether the simulated packet is allowed or blocked; if no rule matches, the configured default policy is applied.

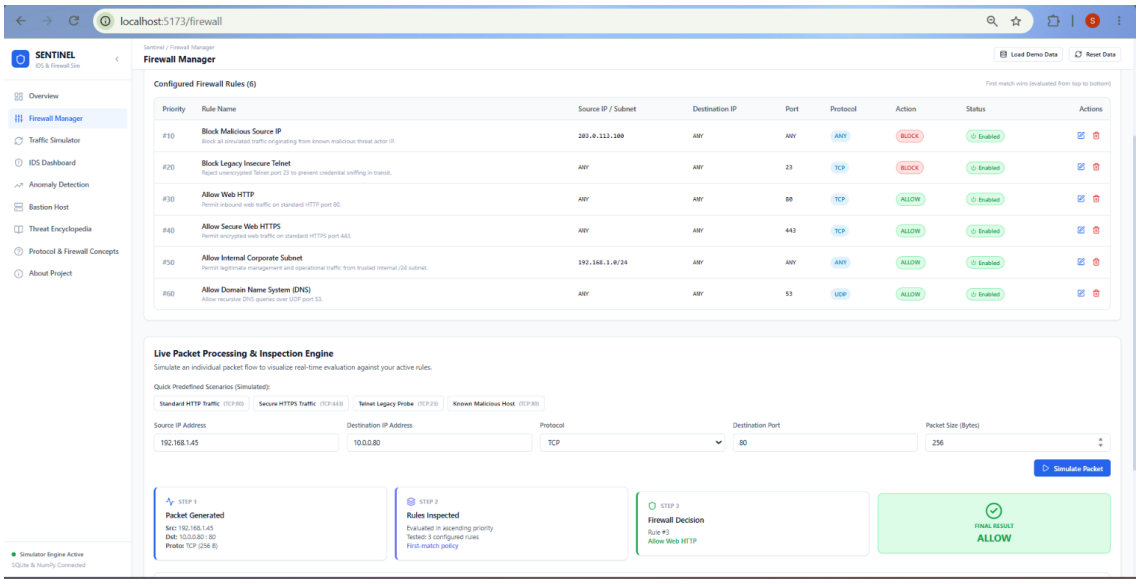


Figure 4.2: Firewall rules and simulated packet evaluation

4.3 Traffic Simulator

The traffic simulator generates predefined or custom simulated packets and records the outcomes. The event log includes source and destination addresses, protocol, port, packet size, firewall decision, matched rule, and IDS status. Example scenarios include failed SSH logins, Telnet probes, port scans, and high-volume traffic.

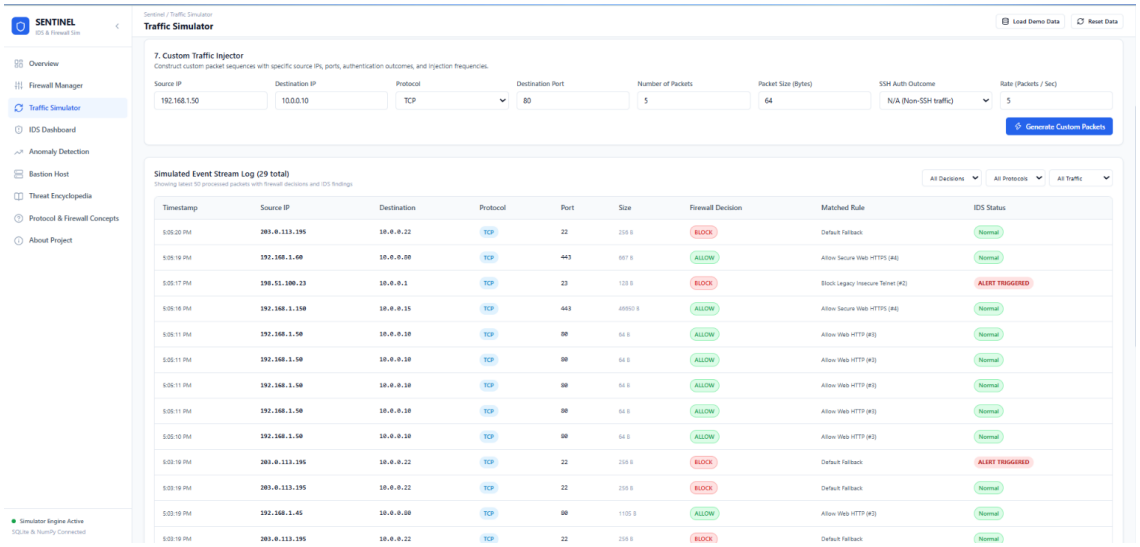


Figure 4.3: Custom traffic generation and simulated event log

4.4 Bastion Host

The bastion-host module illustrates a controlled administrative access path between an untrusted external network and a protected internal network. It includes simulated host configurations and an access-verification tool. The displayed connection status is a simulation, not a real network tunnel.

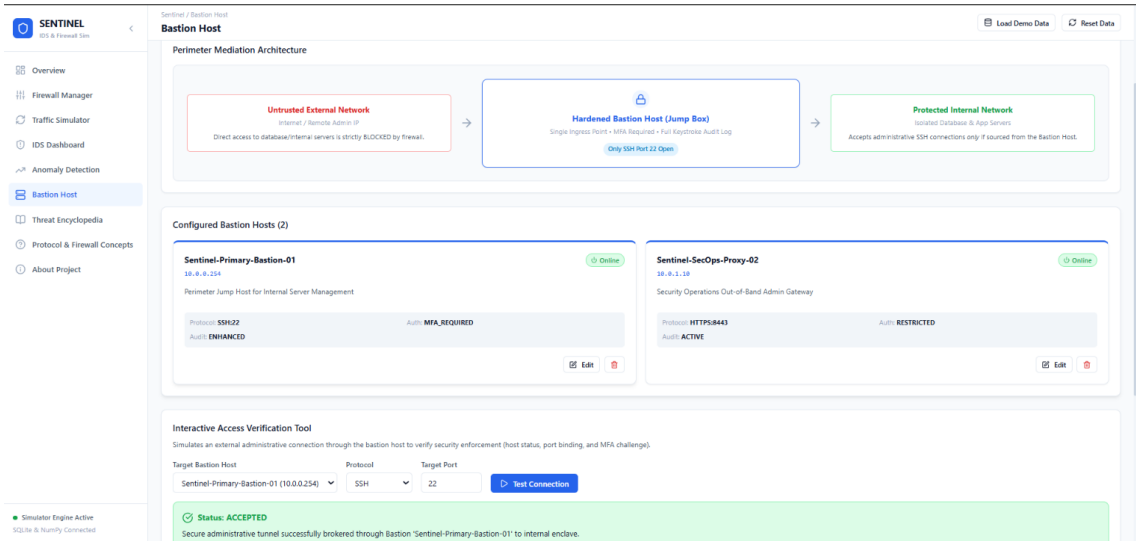


Figure 4.4: Bastion-host architecture and access verification

4.5 Intrusion Detection and Anomaly Detection

SENTINEL contains two detection approaches. The rule-based IDS looks for configured patterns such as repeated failed SSH logins, Telnet connections, port scanning, excessive packet volume, and repeated firewall denials. The statistical anomaly engine compares observed traffic metrics with a historical baseline using the mean, standard deviation, and a configurable Z-score threshold. An anomaly indicates unusual activity and does not, by itself, prove an attack.

4.6 Threat Encyclopedia and Firewall Concepts

The educational pages describe common threats and defensive measures and compare packet-filtering, stateful-inspection, proxy, and next-generation firewalls.

Chapter 5

Implementation and Testing

5.1 SENTINEL Processing Flow

1. Generate a simulated packet or traffic scenario.
2. Validate the packet fields.
3. Evaluate enabled firewall rules by priority.
4. Apply the matching ALLOW or BLOCK action, or the default policy.
5. Store the simulated event in SQLite.
6. Evaluate rule-based IDS signatures.
7. Update statistical traffic metrics and anomaly status.
8. Generate alerts when configured conditions are met.
9. Display the decision, event history, and alerts in the dashboard.

5.2 API and Execution

Both projects use a frontend and backend. The documented local frontend address is http://localhost:5173; the backend API uses http://127.0.0.1:8000, with FastAPI documentation at http://127.0.0.1:8000/docs. SENTINEL's documented startup commands are shown below.

```
cd D:\IASP3\backend
python -m uvicorn app.main:app --reload --host 127.0.0.1 --port
   8000
```

Listing 5.1: SENTINEL backend startup

```
cd D:\IASP3\frontend
npm run dev
```

Listing 5.2: SENTINEL frontend startup

SecureTunnel's startup instructions depend on the local repository path. Use the startup scripts or the frontend and backend commands provided with that project.

5.3 Testing

The supplied SENTINEL documentation reports 12 backend tests passed in 2.75 seconds. These tests cover blocked and allowed traffic, rule priority, disabled rules, SSH and Telnet detections, port scanning, statistical anomalies, bastion-host persistence, rule persistence, input validation, and API integration. The SecureTunnel documentation supplied for this report records 17 backend tests passed and nine frontend test suites passing. These are previously reported results and should be rerun when fresh verification is required.

Chapter 6

Limitations and Conclusion

6.1 Limitations

Both applications are designed for education. SecureTunnel demonstrates selected protocol flows and does not establish real TLS, SSH, or IPsec connections. SENTINEL generates simulated packets and alerts in software; it does not capture physical network traffic, execute live attacks, or modify the operating system firewall. Results should therefore be described as demonstrations rather than production security monitoring.

6.2 Conclusion

SecureTunnel and SENTINEL provide complementary learning tools for Information Assurance and Security. SecureTunnel focuses on understanding secure communication protocols, while SENTINEL focuses on firewall decisions, intrusion signatures, traffic anomalies, and bastion-host access control. Together, they present core network-security concepts through interactive interfaces, visual feedback, and documented back-end logic.

6.3 Future Enhancements

Possible improvements include adding more guided tutorials, expanding test coverage, improving accessibility, exporting reports, and clearly distinguishing simulated behavior from real cryptographic or network operations.

GitHub Repository

The source code and project files for this work are available in the following GitHub repository:

Repository:

https://github.com/satwika06bhaskar-design/IAS-2420030172/tree/main

References

1. SecureTunnel project README and prototype documentation supplied for this report.

2. SENTINEL: Intrusion Detection and Firewall Simulator, system and technical documentation supplied for this report.

3. Rescorla, E., *The Transport Layer Security (TLS) Protocol Version 1.3*, RFC 8446, IETF.

4. Ylonen, T. and Lonvick, C., *The Secure Shell (SSH) Protocol Architecture*, RFC 4251, IETF.

5. Kent, S. and Seo, K., *Security Architecture for the Internet Protocol*, RFC 4301, IETF.